3.5 profit max

DEDEC
DEECB
DBAAD
BDC

1. Assume that a firm is maximizing short-run profits and that price is greater than average variable cost. Which of the following must be true at the firm's level of output?

(A) Marginal revenue is equal to average total cost.
(B) Marginal revenue is greater than total variable cost.
(C) Marginal revenue is equal to price, which is greater than average total cost.
(D) Marginal revenue is equal to marginal cost. ✓
(E) Price is equal to average total cost.

2. At a firm's current rate of output, the marginal cost is $65, the average variable cost is $35, the average fixed cost is $30, and the product price is $65. Which of the following statements is true for the firm?

(A) Economic profits are zero because marginal revenue equals marginal cost.
(B) Economic profits are negative because total revenue is less than total cost.
(C) Economic profits are positive because total revenue is greater than total cost.
(D) Economic profits are negative because price is greater than average variable cost.
(E) Economic profits are zero because price equals average total cost. ✓

3. At the current output level, a firm finds that it has the potential to increase its profit by expanding output. If P = price, MR = marginal revenue, and MC = marginal cost, which of the following must hold at the current output for this firm?

(A) P = MR < MC
(B) P = MR = MC
(C) MR = MC
(D) MR > MC ✓
(E) MR < MC

The chart below gives a firm's total cost of producing different levels of output.

Output	Total Cost
0	$13
1	20
2	25
3	28
4	32
5	43
6	60

4. The profit-maximizing level of output for this firm is

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(A) 2
(B) 3
(C) 4
(D) 5
(E) impossible to determine from the information given ✓

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5. A competitive profit-maximizing firm is currently producing at an output level at which the marginal revenue is equal to marginal cost. Which of the following changes will NOT affect the profit-maximizing quantity?

(A) An increase in the price
(B) An increase in variable costs
(C) An increase in fixed costs ✓
(D) An increase in labor productivity
(E) An increase in the firm's revenues

Answer C

Correct. An increase fixed costs will increase the firm's total cost but not marginal cost. Marginal cost only changes with changes in variable cost. Therefore, marginal revenue will continue to be equal to marginal cost and the firm will not change its quantity to maximize its profit.

The following questions are based on the table below, which gives cost information for a perfectly competitive firm.

Quantity	Average Fixed Costs	Average Variable Costs	Marginal Costs
0			
1	$100.00	$55.00	$55.00
2	50.00	45.00	35.00
3	33.33	50.00	60.00
4	25.00	55.00	70.00
5	20.00	60.00	80.00
6	16.67	65.00	90.00

6. If the product price is $85, how many units of output must the firm produce in order to maximize profits?

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(A) 0
(B) 3
(C) 4
(D) 5 ✓
(E) 6

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7. Greencom is currently producing 20 units of output. At this level of output, Greencom's marginal revenue is $6, marginal cost is $6, and average total cost is $9. If Greencom operates in a monopolistically competitive market and sells all 20 units of output for $10 per unit, it can be concluded that

(A) Greencom would earn more profit by decreasing output
(B) Greencom would earn more profit by increasing output
(C) Greencom is earning zero economic profit
(D) Greencom is earning negative economic profit
(E) Greencom is earning positive economic profit ✓

Answer E

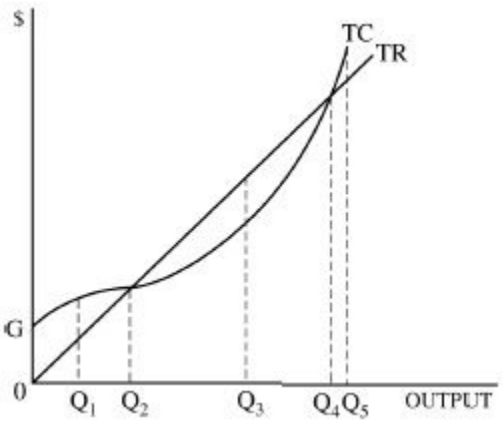
Correct. Greencom is earning positive economic profit, as the firm is earning per-unit profits of $1 (price minus average total cost) multiplied by 20 (the quantity of units sold) for total economic profit of $20.

8. If total revenue is increasing as output increases, marginal revenue is always

(A) equal to average revenue
(B) less than average revenue
(C) increasing
(D) decreasing
(E) greater than zero ✓

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9. In the graph below, TC is total cost and TR is total revenue.



At which level of output is profit maximized?

(A) Q_1
(B) Q_2
(C) Q_3 ✓
(D) Q_4
(E) Q_5

Quantity of Output	Total Revenue	Total Cost
0	$0	$10
1	8	13
2	16	14
3	24	17
4	32	22
5	40	29
6	48	38
7	56	49

10. The marginal revenue of the third unit of output is

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(A) $3

(B) $8 ✓

(C) $10

(D) $17

(E) $24

11. The firm's profit maximizing quantity is

(A) 2

(B) 3

(C) 4

(D) 5 ✓

(E) 6

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12. The most profitable level of output for any firm operating in the short run is the level of output at which

(A) marginal revenue exceeds marginal cost by the highest amount

(B) marginal revenue equals marginal cost ✓

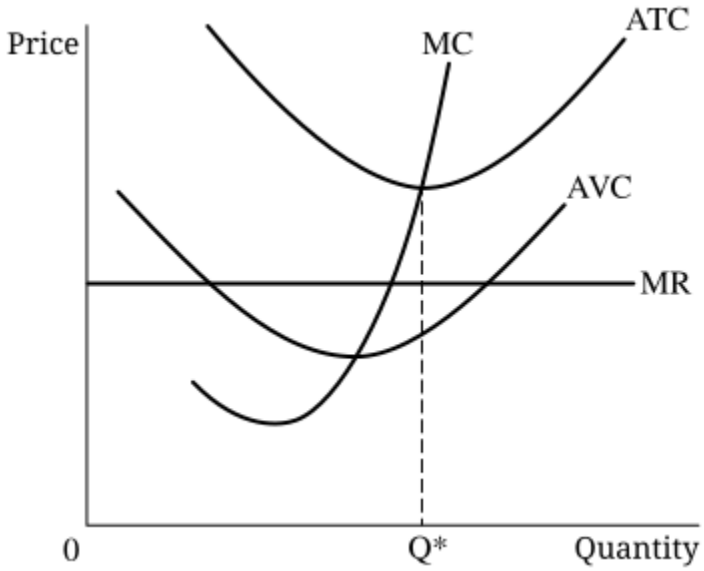
(C) price exceeds average cost by the highest amount

(D) price equals average cost

(E) price equals marginal cost

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13.



The graph provided shows the marginal revenue (MR), average variable cost (AVC), average total cost (ATC), and marginal cost (MC) curves for a perfectly competitive firm. Which of the following is true if the firm is currently producing Q* units of output?

(A) The firm can increase its economic profit by producing fewer units of output. ✓

(B) The firm can increase its economic profit by increasing the price of its product.

(C) The firm is maximizing its economic profit at the current level of output.

(D) The firm can increase total revenue by producing fewer units of output.

(E) The firm can increase total revenue by decreasing the price of its product.

Answer A

Correct. Firms maximize economic profit by producing the quantity of output where marginal revenue equals marginal cost. At Q*, marginal cost is greater than marginal revenue; therefore, the firm can increase its economic profit by producing fewer units of output until marginal revenue equals marginal cost.

14. Assume a competitive firm is producing where price (P) and marginal revenue (MR) are greater than marginal cost (MC) and average variable cost (AVC). Which of the following is true regarding the firm's short-run output level?

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(A) The firm is producing too little and should increase its output level until $\mathrm{P} = \mathrm{MR} = \mathrm{MC}$. ✓

(B) The firm is producing too much and should reduce its output level until $\mathrm{P} = \mathrm{MR} = \mathrm{AVC}$.

(C) The firm is not maximizing profits and should raise its price but not change its output level.

(D) The firm should increase its marginal revenue to equal price and reduce its output level until $\mathrm{MR} = \mathrm{MC}$.

(E) The firm should reduce its price until $\mathrm{P} = \mathrm{MR} = \mathrm{MC}$.

Answer A

Correct. The competitive firm should continue to produce up to the output level where price, which is equal to marginal revenue (MR) for a competitive firm, equals marginal cost (MC). If P and MR exceed MC, then there are units of output that the firm is not producing that would be profitable.

15. When a competitive firm maximizes short-run economic profits, it produces at the output level where

(A) average cost is at its minimum

(B) price equals zero

(C) price equals average variable cost

(D) marginal revenue equals marginal cost ✓

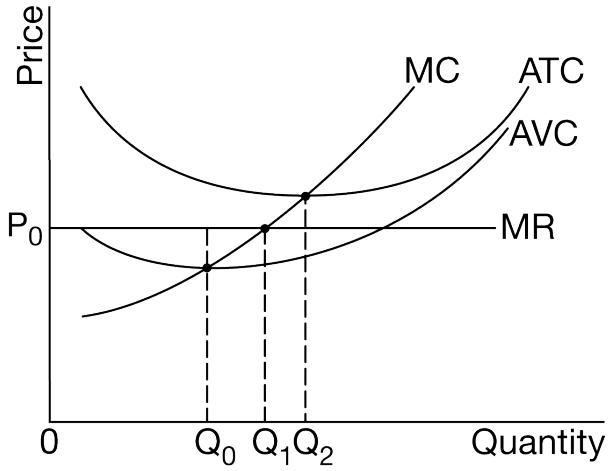
(E) marginal revenue equals average variable cost

Answer D

Correct. Profit maximization occurs where marginal revenue (price) equals marginal cost.

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16.



The graph above shows the short-run cost and revenue curves for a perfectly competitive firm. Assume that the market price is P_0 and the firm is producing at quantity Q_2. To maximize profit, the firm should

(A) continue to produce quantity Q_2, where average total cost is at its minimum

(B) produce quantity Q_1, where price is equal to marginal cost ✓

(C) produce quantity Q_0, where average variable cost is at its minimum

(D) decrease the price so that price equals average variable cost

(E) increase the market price to the level of the minimum average total cost

Answer B

Correct. The profit-maximizing competitive firm should produce where P (or MR) $=$ MC at Q_1. The firm will have a loss, since P is less than ATC.

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17.

Quantity of Output (units)	Total Variable Cost ($)	Total Cost ($)
1	100	200
2	190	290
3	270	370
4	340	440
5	420	520
6	510	610
7	610	710

The table provided shows the total variable cost and total cost of production for varying quantities of output by a typical profit-maximizing firm in a perfectly competitive market. How many units will the firm produce at a market price of $95 to maximize profits?

(A) 2 units

(B) 3 units

(C) 4 units

(D) 6 units ✓

(E) 7 units

Answer D

Correct. To maximize profits (or minimize losses), firms will produce up to the last unit where marginal revenue is greater than or equal to marginal cost. This occurs at 6 units of output, where the change in variable cost (marginal cost) is closest to marginal revenue, which is the market price ($90 and $95, respectively) and price is still greater than marginal cost.

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18.

Quantity (units)	Total Revenue ($)	Total Cost ($)
0	0	100
1	200	150
2	250	190
3	275	210
4	290	225
5	300	255
6	305	300

In the table provided, at which quantity is the marginal revenue equal to marginal cost?

(A) 2
(B) 3
(C) 4 ✓
(D) 5
(E) 6

Answer C

Correct. Marginal revenue is the change in total revenue divided by the change in quantity, and marginal cost is the change in total cost divided by the change in quantity. Marginal revenue is equal to marginal cost at quantity of 4 units. Marginal revenue $= \frac{\$290-\$275}{4-3} = \$15$ and marginal cost $= \frac{\$225-\$210}{4-3} = \$15$.